



CSC 482

Final Project



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Siri Gunturi, and Ellie Pearson

Team

Harini Baskar

- Worked on rule-based extractor

Kaitlyn Carrillo

- Worked on parser and report

Siri Gunturi

- Worked on data collection and visualization

Ellie Pearson

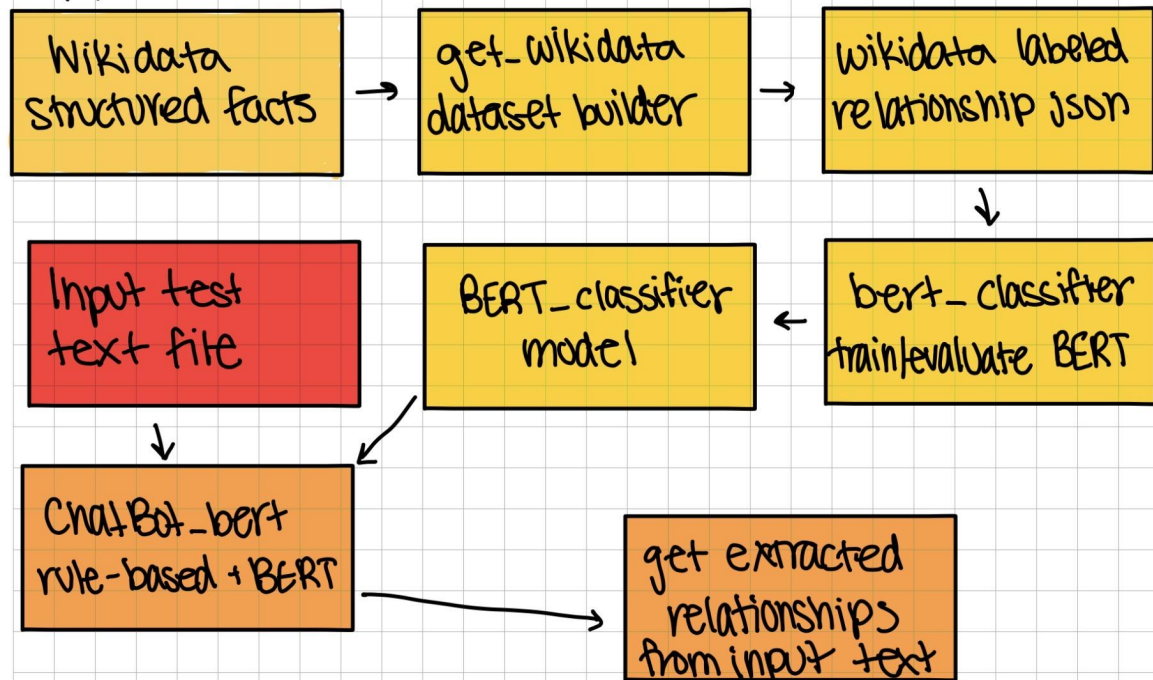
- Worked on bert classifier



Genealogy Extractor



Full pipeline





Demo



Technical Achievements

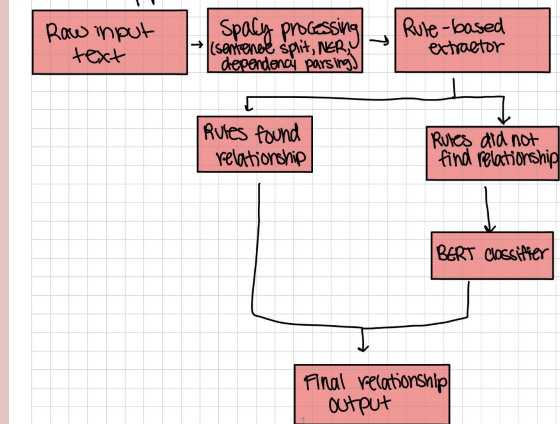
Developed by team:

Rule-based extractor,
dataset generation
process, Bert-based
relational classifier

APIs: Wikipedia API

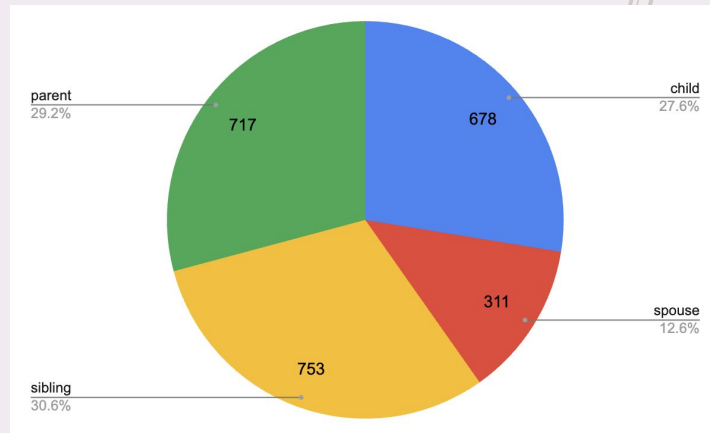
```
{  
  "text": "From then on, Anne, [  
  "input_text": "From then on, [  
  "person": "Anne Frank",  
  "relative": "Margot Frank",  
  "person_mention": "Anne",  
  "relative_mention": "Margot",  
  "relation": "sibling"  
},
```

ChatBot-bert pipeline



Constructing the Dataset

1. Get relationships and wikipedia article from API
2. Clean wikitext
3. Parse sentences for mentions of both subject and relative
4. Tag sentences to indicate people for BERT model



Classifier

1. BERT classifier to capture context of sentence
2. Cross fold validation with 5 folds and 3 epochs
3. Struggles the most with child

```
===== Held-out test results =====  
Test accuracy: 0.7800  
Test macro-F1: 0.7312
```

	precision	recall	f1-score	support
parent	0.9176	0.8298	0.8715	94
child	0.7143	0.5128	0.5970	39
spouse	0.5143	0.8182	0.6316	22
sibling	0.7692	0.8889	0.8247	45
accuracy			0.7800	200
macro avg	0.7289	0.7624	0.7312	200
weighted avg	0.8002	0.7800	0.7811	200



Rule-Based Extractor

1. Uses a combination of spaCy dependency parsing and NER.
2. Using the parser we could successfully extract direct relationships and their inverses (parent ↔ child), providing a structured view of familial links.

```
(base) harinibaskar@Harinis-MacBook-Air-551 Project % python3 chatBot.py test.txt
Augustine → parent → George Washington
George Washington → child → Augustine
George Washington → child → Mary Ball Washington
George Washington → sibling → Lawrence
Lawrence → sibling → George Washington
Mary Ball Washington → parent → George Washington
```




Data & Visualization

```
[sirigunturi@Siris-MacBook-Pro-2 Project % python3 chatBot_bert.py test/hughes_in.txt  
✗ spaCy model en_core_web_lg version 3.8.0 is not supported by  
Coreferee. Please examine /coreferee/lang/en/config.cfg to see the supported  
models/versions.
```

Rule Based Extraction

Jim Hughes — Ellen Weinberg-Hughes

Jack Hughes + Luke + Quinn

Results



	precision	recall	f1-score	support
parent	0.7481	0.8145	0.7799	124
child	0.8333	0.6667	0.7407	120
spouse	0.7561	0.7209	0.7381	43
sibling	0.7985	0.8992	0.8458	119
accuracy		0.7857		406
macro avg	0.7840	0.7753	0.7762	406
weighted avg	0.7889	0.7857	0.7832	406





Limitations

01.

Misidentification



02.

Code Computation Time



03.

Text Ambiguity





Future Possibilities



01.

Type of Dataset

Incorporate human-annotated datasets to increase accuracy and reduce errors.

02.

LLMs

Integrate LLMs to validate, correct, or infer missing relationships.

03.

More data

Expand our dataset